Urban Heat Island Analysis of Pune

1. INTRODUCTION:

Urban Heat Island (UHI) refers to the phenomenon where urban areas experience higher temperatures than surrounding rural regions due to increased built-up surfaces, anthropogenic activities, and reduced vegetation cover. Materials such as concrete and asphalt absorb and retain heat, resulting in elevated surface temperatures in urban environments.

Understanding UHI is important for urban planning, environmental sustainability, and improving living conditions. This study analyses the relationship between vegetation cover and land surface temperature using geospatial techniques.

2. OBJECTIVES:

- To derive Normalized Difference Vegetation Index (NDVI) from satellite imagery
- To calculate Land Surface Temperature (LST)
- To identify Urban Heat Island zones
- To analyse the relationship between vegetation and temperature

3. DATA USED

Satellite Data: Landsat 8 imagery (USGS)

Bands Used:

- Red and Near Infrared (for NDVI)
- Thermal Infrared Band (Band 10) for LST

Study Area Boundary: Pune administrative boundary

Projection: UTM (WGS 84)

4. Methodology

4.1 NDVI Calculation

NDVI was calculated using the formula:

NDVI = (NIR - Red) / (NIR + Red)

NDVI values range from -1 to +1, where higher values indicate dense vegetation and lower values indicate built-up or barren areas.

4.2 Land Surface Temperature (LST)

Land Surface Temperature (LST) was derived from the thermal infrared band (Band 10) of Landsat 8 imagery. Calibration constants including radiance multiplicative and additive factors, along with thermal constants (K1 and K2), were obtained from the metadata file (MTL.txt).

The process involved converting digital numbers into spectral radiance, followed by transformation into brightness temperature using standard equations. The final temperature values were obtained in degrees Celsius.

4.3 Reclassification

- NDVI was classified into vegetation categories
- LST was classified into temperature zones
- Both layers were used to generate UHI map

4.4 Urban Heat Island Mapping

UHI was derived using raster analysis:

UHI = LST_class - NDVI_class

This helped identify zones of heat concentration and cooling areas.

4.5 Correlation Analysis

A scatter plot between NDVI and LST values was generated using sampled raster data to analyze their relationship.

5. Results and Discussion

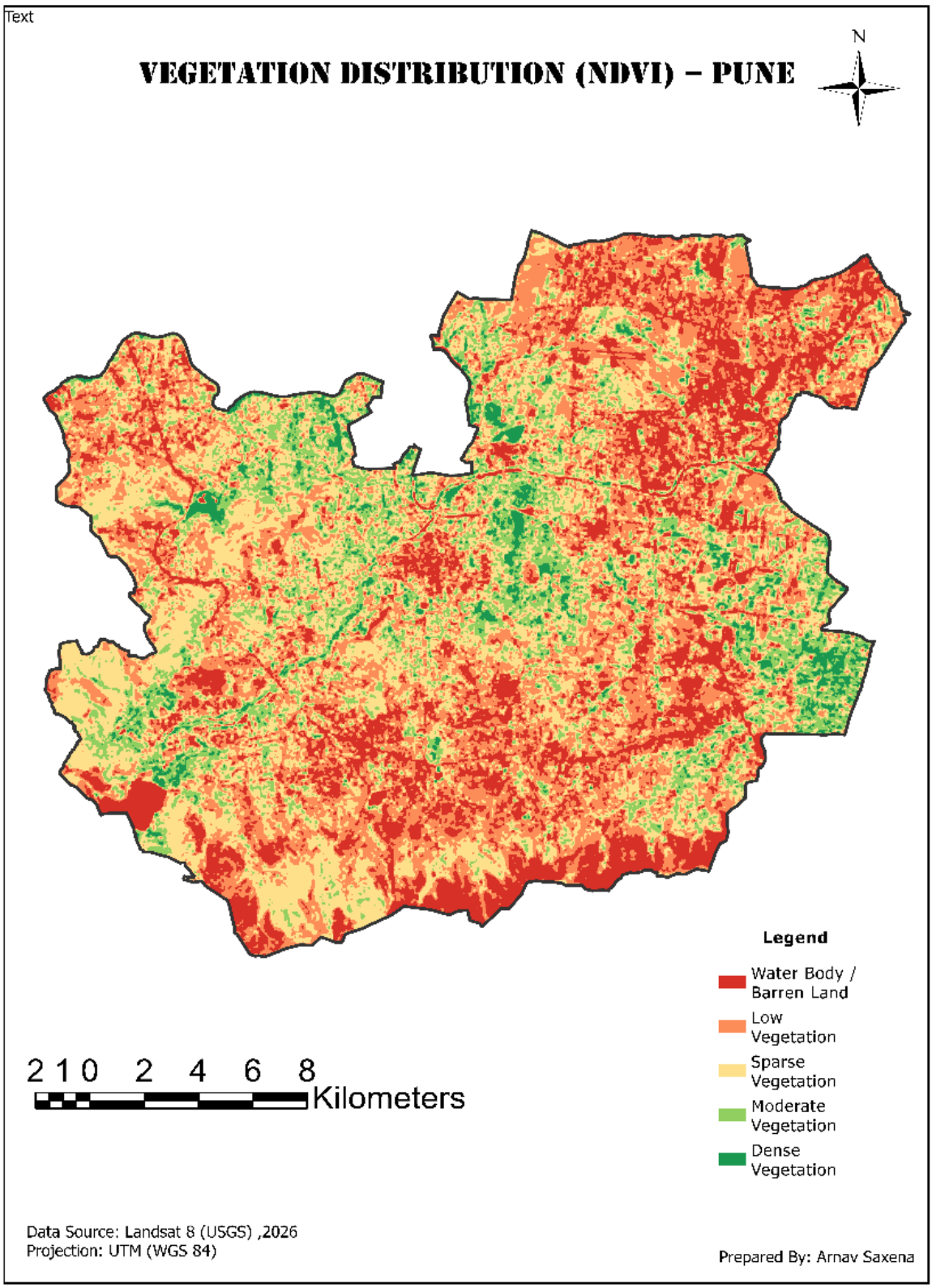

Figure 1. NDVI Map of Pune (Source: Author,2026)

1. NDVI Map

The NDVI map indicates a heterogeneous distribution of vegetation across the study area. Higher NDVI values are predominantly observed in peripheral and less urbanized regions, suggesting dense vegetation cover. In contrast, central urban areas exhibit lower NDVI values, indicating limited green cover and dominance of built-up surfaces.

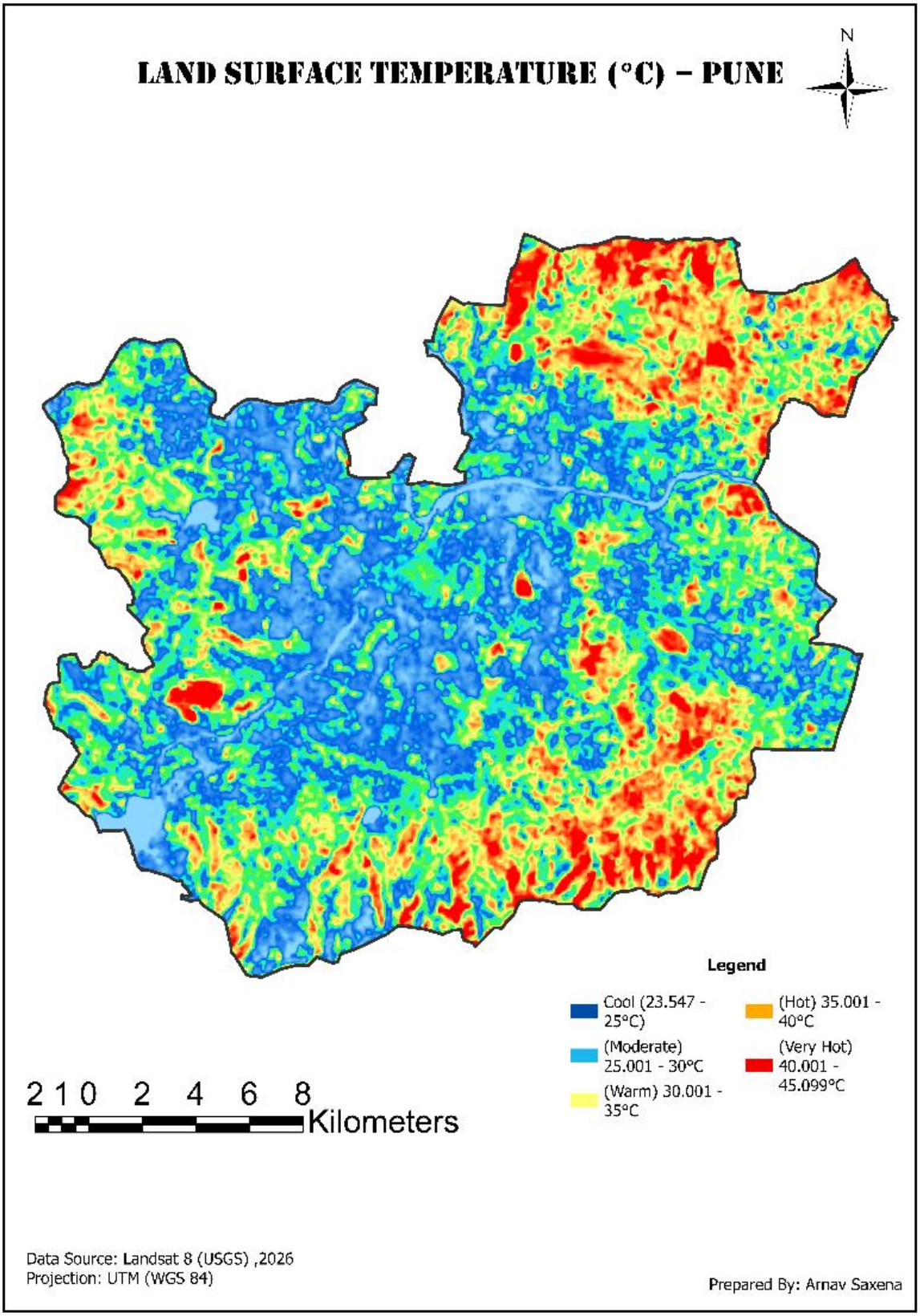

Figure 2. LST Map of Pune (Source: Author, 2026)

2. LST Map

The Land Surface Temperature map reveals significant spatial variation across Pune. Higher temperature zones are concentrated in densely built-up areas, while relatively lower temperatures are observed in vegetated and water-covered regions. This highlights the influence of land cover on surface thermal characteristics.

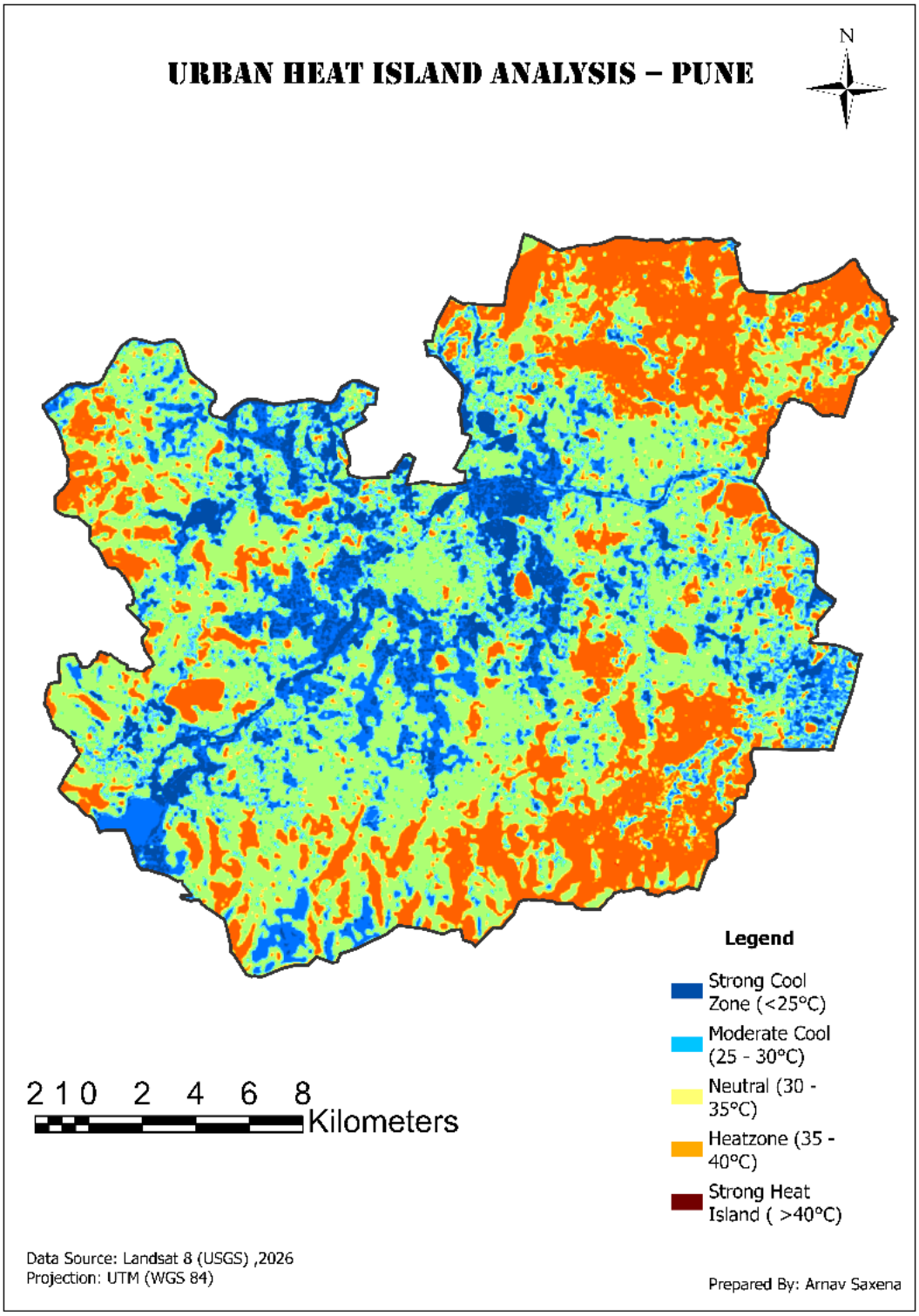

Figure 3. UHI Map of Pune

UHI Map

The UHI map clearly delineates zones of heat concentration and cooling effects. Urban cores exhibit strong heat island characteristics, whereas vegetated regions function as cooling zones. The spatial pattern demonstrates the direct impact of urbanization on surface temperature distribution.

The spatial pattern clearly indicates that densely built-up areas contribute significantly to heat accumulation, whereas vegetated regions mitigate temperature through evapotranspiration and shading effects.

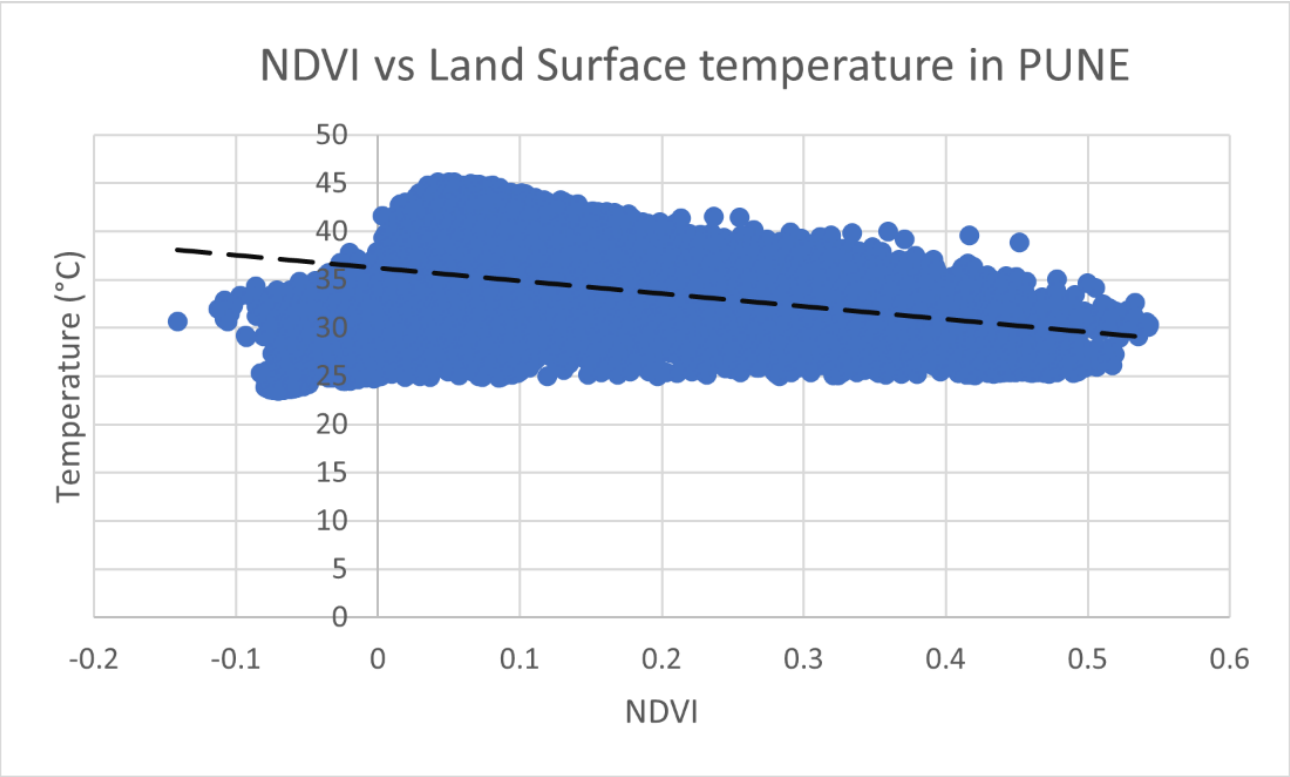

Figure 4. NDVI vs LST Scatter Plot

NDVI vs LST Graph

The scatter plot analysis indicates a negative correlation between NDVI and land surface temperature. Areas with higher vegetation cover tend to exhibit lower temperatures due to evapotranspiration and shading effects. However, the relationship is not strictly linear due to mixed land use patterns, urban materials, and environmental variability.

6. Conclusion

The study confirms that vegetation plays a significant role in regulating land surface temperature. Areas with higher vegetation cover exhibit lower temperatures, while built-up regions contribute to increased heat accumulation. These findings highlight the importance of incorporating green infrastructure in urban planning to mitigate the Urban Heat Island effect and improve environmental sustainability. This pattern highlights the strong influence of land use and land cover on urban thermal dynamics.

This analysis demonstrates the practical application of geospatial techniques in understanding urban environmental challenges.

7. Tools and Software Used

- ArcGIS Pro
- Remote Sensing techniques
- Microsoft Excel